

Debabrata Sahu

AI/ML Engineer | Computer Vision & Deep Learning Developer
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GitHub: <https://github.com/Deba-0075>



OBJECTIVE

AI/ML Engineer specializing in Computer Vision and Deep Learning with experience in building real-time AI applications using Python, OpenCV, TensorFlow, and Machine Learning techniques. Skilled in developing intelligent solutions for healthcare, accessibility, and automation using data preprocessing, model training, and deployment with Flask and Streamlit.

TECHNICAL SKILLS

Programming Language: Python, C

Machine Learning & AI: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Predictive Modeling.

Deep Learning: CNN, Neural Networks, TensorFlow, PyTorch, Image Classification, Transfer Learning.

Computer Vision: OpenCV, MediaPipe, Image Processing, Object Detection, Image Segmentation, Feature Extraction.

Data Science & Analytics: NumPy, Pandas, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Data Visualization.

Frameworks & Tools: Scikit-learn, Flask, Streamlit, REST API, Git, GitHub, Jupyter Notebook, VS Code

Database & Platforms: MySQL, Windows, GCP, Vertex AI, BigQuery

Currently Exploring: Generative AI, RAG (Retrieval-Augmented Generation), Transformers, LangChain, YOLO, Edge AI

PROJECT EXPERIENCE

➤ Bi-Directional Sign Language to Speech Interpreter

Tech Stack: Python, OpenCV, MediaPipe, CNN, SpeechRecognition, DeepFace, gTTS

- Developed a real-time AI system to convert sign language gestures into text and speech using computer vision and deep learning.
- Trained CNN models on hand landmark features for accurate gesture recognition and integrated speech synthesis modules.
- Achieved high accuracy in real-time classification and won 1st Prize at BPUT Hackathon 2024 (Accessibility Track).
- **GitHub:** <https://github.com/Deba-0075/Sign-to-Speech>

➤ AI-Driven Solar Panel Cleaning System

Tech Stack: Python, Computer Vision, IoT, Sensors, Automation

- Developed an IoT-based intelligent system to detect dust accumulation on solar panels using computer vision techniques.
- Designed and integrated a sensor-based automated cleaning mechanism to improve energy efficiency and reduce manual maintenance.
- Combined hardware and software components for real-time monitoring and automated decision-making.
- **LinkedIn:** <https://www.linkedin.com/in/debabrata-sahu-293ba0304/>

➤ Camouflage Detection System

Tech Stack: Python, OpenCV, Machine Learning

- Designed a computer vision model to detect camouflaged objects using image segmentation and feature extraction on the COD10K dataset.
- Applied preprocessing and feature engineering techniques to improve detection in low-contrast environments.
- Achieved 89% accuracy in challenging visual conditions.
- **GitHub:** https://github.com/Deba-0075/Camouflaged_Object_Detection

➤ **Eye Disease Detection System**

Tech Stack: Python, CNN, OpenCV, Medical Image Processing

- Developed a deep learning model using CNN to detect eye diseases from retinal fundus images.
- Performed data preprocessing, augmentation, and model optimization on 4,000+ images.
- Achieved 92% accuracy, supporting early diagnosis in healthcare applications.
- **GitHub:** <https://github.com/Deba-0075/Eye-Disease-Detection>

➤ **COVID-19 Detection using Blood Reports**

Tech Stack: Python, Machine Learning, Healthcare Data

- Built a machine learning model to predict COVID-19 risk using hematological parameters from 70,000+ patient samples.
- Applied data cleaning, feature engineering, and model evaluation techniques to improve prediction performance.
- Achieved 97% accuracy, enabling effective early detection.
- **GitHub:** <https://github.com/Deba-0075/covid19-detection-ml>

ACHIEVEMENTS AND PUBLICATIONS

• **Two-Time Winner – BPUT Hackathon (2024 & 2025)**

Built AI-based solutions in accessibility, sustainability, and automation.

• **Research Publication** – “Analysis on Deep Architectures for Automatic Modulation Classification”
Published and presented at Roland Institute of Technology, Berhampur (Jan 2025).

• **Research Publication** – “Adaptive Compression for Dynamic Agricultural Environments: A Theoretical Framework for Real-Time, Resource-Aware Deep Learning on Edge AI”
Presented at 2nd International Conference on Green Innovation, Kalam Institute of Technology, Nov 2025.

EDUCATION

B.Tech in Computer Science and Engineering (2022 – 26)

Kalam Institute of Technology, Berhampur | **CGPA:** 8.29

Affiliated to Biju Patnaik University of Technology (BPUT), Rourkela

Senior Secondary (12th) – CHSE, Odisha (2020-22)

Saraswati Vidya Mandir Higher Secondary School, Kalingard, Aska

Secondary (10th) – BSE, Odisha (2019-20)

Uchhabeswar Up graded Govt Nodal High School, Munigadi

CERTIFICATIONS

- **Research Paper Presentation Certificates** – 2 Publications
- **BPUT Hackathon** – Two-Time Winner Certificate (2024 & 2025)
- **NMIET Hackathon** – Participant, Bhubaneswar
- **BPUT Project Exhibition** – Participant
- **Google AI Essentials**
- **Google Cloud Skills Boost**
- **Vertex AI Introduction**
- **Microsoft AI/ML Engineer Professional Certificate (Coursera)**

SOFT SKILLS

Problem Solving, Team Collaboration, Analytical Thinking, Communication, Fast Learner

LANGUAGES KNOWN

English, Hindi, Odia (Native)

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